Restaurant Vibe Classification with Locally Fine-Tuned DistilBERT

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Abstract

This project explores the use of locally fine-tuned transformer models for restaurant vibe classification using Yelp review text and restaurant metadata. The goal was to classify restaurants into one of seven atmosphere-based categories: *study_spot*, *date_night*, *casual_hangout*, *family_friendly*, *upscale*, *trendy*, and *late_night*. A custom dataset was constructed from the Yelp Open Dataset by filtering restaurant businesses located in Santa Barbara, Goleta, and Carpinteria, then merging business metadata with associated review text. Approximately 2,000 labeled examples were manually balanced across the seven vibe categories. The primary model used in this project was DistilBERT fine-tuned locally for multi-class sequence classification using Hugging Face Transformers and PyTorch. Multiple fine-tuning configurations were tested and compared against a traditional NLP baseline using TF-IDF vectorization and Logistic Regression classification. The final DistilBERT model achieved an accuracy of 82.7% and a weighted F1-score of 0.826, outperforming the baseline model, which achieved 70% accuracy and a weighted F1-score of 0.70. Confusion matrix analysis and qualitative prediction examples showed that the transformer model was better able to capture contextual and semantic relationships between overlapping restaurant vibe categories. The results demonstrate that locally fine-tuned transformer models can succeed at performing domain-specific restaurant atmosphere classification and benefit substantially from larger, more diverse labeled datasets compared to traditional NLP approaches.

Keywords: Natural Language Processing, DistilBERT, Transformer Models, Text Classification, Yelp Reviews, Restaurant Recommendation Systems, Fine-Tuning, Sentiment and Atmosphere Analysis

1 Introduction

People, like myself, often use reviews to describe the overall atmosphere or "vibe" of a restaurant, including whether it feels like a good place to study, go on a date, spend time with family, or enjoy nightlife with friends. These descriptions are highly subjective and are usually embedded within long-form natural language reviews, making them difficult to organize or search using traditional restaurant recommendation systems. Users can find it difficult to assess the vibe of a restaurant without allotting time to reading through various reviews themselves.

The original motivation for this project came from wanting a better way to narrow restaurant choices based on the type of experience or atmosphere desired for a meal. While existing restaurant platforms provide filters for cuisine, price range, and ratings, they rarely capture more nuanced information such as whether a restaurant feels trendy, relaxed, upscale, study-friendly, or appropriate for a date night. The broader long-term goal was to eventually build a recommendation assistant capable of combining vibe classification with additional restaurant information such as online menu links, parking availability, estimated wait times, and other useful dining context.

Initially, the project aimed to focus specifically on Orange County restaurants. However, discovering a large, high-quality, publicly available Orange County restaurant review dataset proved difficult without relying on paid commercial datasets. As a result, the project scope shifted toward restaurants located in Santa Barbara, Goleta, and Carpinteria using the freely available Yelp Open Dataset. Narrowing the geographic scope allowed the project to focus more heavily on the NLP classification pipeline, local fine-tuning process, and evaluation methodology.

This project explores whether transformer-based NLP models can classify restaurant vibes using Yelp review text and restaurant metadata. More specifically, the project focuses on fine-tuning DistilBERT locally for a multi-class restaurant vibe classification task. The model was trained and evaluated entirely on my laptop's local hardware using Hugging Face Transformers and PyTorch. The training and fine-tuning discusssed in this paper do not rely on any cloud-based training infrastructure.

To evaluate the effectiveness of transformer fine-tuning, the project also compares DistilBERT against a traditional NLP baseline using TF-IDF vectorization and Logistic Regression classification. Multiple fine-tuning configurations were tested to compare learning rates, warmup scheduling, and training duration. The final system was evaluated using quantitative metrics such as accuracy, precision, recall, and weighted F1-score, along with qualitative prediction analysis and confusion matrix evaluation.

Overall, this project demonstrates how locally fine-tuned transformer models can be applied to a focused domain-specific NLP task involving restaurant atmosphere classification and contextual understanding of user-generated review text, while also serving as a foundation for future restaurant recommendation and discovery systems.

2 Problem Definition

This project frames restaurant vibe prediction as a supervised text classification task. The goal is to classify a restaurant into one of several "vibe" categories using Yelp review text and restaurant metadata.

The input to the model consists of the following fields combined into a single sequence for training and inference:

- restaurant category metadata
- customer review text

The output of the model is one of seven predefined vibe labels:

Label	Description
study_spot	Quiet cafes and work-friendly spaces
late_night	Bars, nightlife, and evening atmosphere
casual_hangout	Relaxed and informal dining
family_friendly	Restaurants suited for families and groups
upscale	Elegant or fine dining experiences
trendy	Modern, stylish, or social restaurants
date_night	Romantic or intimate dining atmosphere

Table 1: Restaurant vibe classification labels

The task was designed to evaluate whether transformer-based NLP models could learn meaningful contextual patterns related to restaurant atmosphere and dining experiences from unstructured review text.

3 Dataset

3.1 Data Source

The dataset used in this project was constructed using the Yelp Open Dataset, which contains large-scale Yelp business and review data in JSON format. The dataset includes restaurant metadata, user reviews, ratings, and category information for businesses across multiple regions.

The project specifically used information from the `business.json` and `review.json` dataset files. The `business.json` file contained information about the restaurant or business such as restaurant name, categories, city, state, ratings, and review counts. The `review.json` file contained the natural language text from user reviews.

Initially, the project aimed to use a smaller Orange County restaurant dataset obtained through Kaggle because it contained more directly relevant atmosphere-related metadata for each food establishment and required less pre-processing. However, access to the full dataset required a paid license which made its retrieval impractical for this project. As a result, the project shifted towards the use of the large, publically available Yelp Open Dataset and required additional data cleaning and preprocessing to construct a usable dataset for training.

3.2 Dataset Filtering

The Yelp Open Dataset includes businesses from many domains beyond restaurant, including grocery stores, gas stations, hair salons, etc. Because of this, filtering was required before the dataset could be used for restaurant-related vibe classification.

The first stage of filtering focused on selecting restaurant-related businesses through the `categories` field. All unique category values were observed and sorted to determine which labels would most likely apply to a restaurant-related business. Businesses were then retained only if at least one of their category labels was included in the accepted labels list. Some of the labels include: ‘Restaurants’, ‘Breakfast and Brunch’, ‘Pizza’, ‘Cocktail Bars’, ‘Specialty Food’, ‘Sandwiches’, ‘Seafood’. Businesses with missing categories or clearly unrelated business types were removed from the dataset.

The second stage was additional filtering to remove businesses that snuck by with at least of their labels meeting the previous criteria but their other labels indicating they lie outside our area of interest. A list of categories to exclude was constructed and used to remove any businesses that included the labels. Some examples of these labels are 'Tobacco Shops', 'Gas Statations', and 'Adult Entertainment'.

The third stage focused on geographic filtering to only retain businesses located in the Santa Barbara region. Only those noted to be in the cities of Santa Barbara, Goleta, and Carpinteria were kept. This geographic narrowing helped reduce the dataset size while also creating a more locally consistent restuarant domain for experimentation and labeling.

Additional filtering steps removed sparse or low-quality entries, including businesses with missing review text or insufficient metadata. The resulting filtered dataset contained restaurant businesses paired with authentic Yelp review language suitable for downstream NLP classification.

3.3 Dataset Construction

After filtering, the restaurant businesse metadata was merged with the Yelp review dataset using the unique, shared `business_id` field. This created a combined dataset where each row represented an restuarant-review pair. For NLP training, the restaurant categories and review text were combined into a single input sequence. This allowed the model to learn from both the resturant category labels and review text.

The final NLP input format followed the below structure:

```
Categories: Coffee & Tea, Cafes
Review: Quite cafe with students working on their laptops and outlets available
for a couple of seats. I recommend the Ube Latte!
```

3.4 Labeling Process

One of the largest challenges in the project was constructing a balanced and meaningful labeled dataset for restaurant vibe classification. Because the Yelp Open Dataset does not contain predefined atmosphere or 'vibe' labels, a custom labeling pipeline had to be developed manually. The initial approach involved manually labeling a small subset of approximately 200 restaurant-review examples using the seven predefined vibe categories listed in Table 1.

Special attention was given to maintaining balanced label representation across all vibe categories. Early labeling attempts produced strong class imbalance issues because some restaurant vibes naturally appeared more frequently than others within the Yelp review data. To reduce this problem, the dataset was intentionally balanced so that each vibe category contained approximately equal representation during training and evaluation.

After manually labeling the initial 200-example dataset, the process was passed to an AI-assistant to label an additional 1,800 examples in order to create the final 2,000-row dataset used for model training. To accelerate the labeling workflow, AI-assisted label recommendations were used to generate candidate vibe labels for the additional examples. These labels were not accepted blindly; instead, random samples of the generated labels were manually reviewed and corrected when necessary to maintain consistency and reduce labeling errors. Final label counts were checked again after the expanded labeling process to confirm that the dataset remained reasonably balanced across all seven vibe categories.

3.5 Train / Validation / Test Split

The final 2,000-example dataset was divided into training, validation, and test sets using a 70/15/15 split. Stratified splitting was used to preserve balanced label representation across all seven vibe categories within each dataset partition.

The training set was used for model fine-tuning, the validation set was used for monitoring model performance during training, and the test set was reserved for final evaluation and baseline comparison.

4 Model and Fine-Tuning Method

4.1 Model Selection

DistilBERT was selected as the primary model for this project because it provides a strong balance between NLP performance and computational efficiency. DistilBERT is a compressed version of BERT that retains much of the original transformer model's language understanding capability while using fewer parameters and requiring less computational power.

The model was trained and fine-tuned locally on consumer laptop hardware rather than relying on cloud-based GPUs. Larger transformer models would have significantly increased training time and memory requirements, making local experimentation more difficult. DistilBERT was well-suited for this task because it is lightweight enough to support local fine-tuning while still providing strong contextual language understanding for multi-class text classification tasks. The model was trained locally using PyTorch and Hugging Face Transformers without requiring parameter-efficient fine-tuning methods such as LoRA or QLoRA.

4.2 Tokenization

The `DistilBertTokenizer`, provided by the Hugging Face Transformers library, was used to convert restaurant review text into tokenized input sequences compatible with DistilBERT.

Before training, the combined restaurant metadata and review text were tokenized using passing, truncation, and fixed maximum sequence length. Padding ensures that all input sequences had consistent length during batch training. Truncation prevents excessively long reviews from exceeding the model input limit. A maximum sequence length of 128 tokens was applied.

4.3 Fine-Tuning Method

The `DistilBertForSequenceClassification`, also from the Hugging Face Transformers library, was used so that the pretrained DistilBERT model could predict one of the seven restaurant vibe labels.

Fine-tuning was done locally using PyTorch and the Hugging Face `Trainer` API. During training, the model learned from the combined restaurant categories and Yelp review text created during preprocessing. Full fine-tuning was performed rather than parameter-efficient approaches such as LoRA or QLoRA, so all trainable DistilBERT parameters were updated during training. Since DistilBERT is much smaller than larger transformer models, full fine-tuning was still manageable on local laptop hardware.

4.4 Baseline Model

To compare transformer performance against a simpler NLP approach, a traditional machine learning baseline was also implemented using TF-IDF vectorization and Logistic Regression classification. The TF-IDF step converted review text into numerical feature vectors based on word importance and frequency across the dataset. These vectors were then used to train a Logistic Regression classifier for restaurant vibe prediction.

This baseline was useful because it provided a comparison against a more traditional bag-of-words NLP pipeline that relies mostly on keyword patterns instead of contextual language understanding. Comparing the baseline against DistilBERT helped evaluate whether transformer fine-tuning actually improved performance on restaurant vibe classification.

5 Experimental Setup

All preprocessing, training, and evaluation were performed locally using Python and a local virtual environment. The project used several NLP and machine learning libraries throughout development, including: PyTorch, Hugging Face Transformers, scikit-learn, pandas, matplotlib. PyTorch and Hugging Face Transformers were used for DistilBERT fine-tuning, while scikit-learn was used for the TF-IDF baseline model, train/test splitting, and evaluation metrics. Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis.

5.1 Training Configurations

Three DistilBERT fine-tuning configurations were tested to compare learning rate behavior and overall training performance.

Experiment	Learning Rate	Epochs	Notes
Model 1	3e-5	4	Baseline fine-tuning configuration
Model 2	2e-5	4	Lower learning rate with warmup scheduling
Model 3	2e-5	4	Lower learning rate without warmup scheduling

Table 2: DistilBERT experiment configurations

Model 1 achieved the strongest overall test performance and was selected as the final model for comparison against the TF-IDF baseline.

6 Results

Three DistilBERT fine-tuning configurations were tested to compare learning rates, warmup scheduling, and overall model performance. Model 1 achieved the strongest overall performance and generalization on the test set. Model 3 achieved slightly strong validation metrics during training, but Model 1 produced better final test performance, which suggests a stronger generalization to unseen restaurant reviews. Model 2 consistently underperformed relative to the other two models, perhaps due to the combination of a lower learning rate and warmup scheduling, which appeared to slow convergenve too aggressively for the dataset size and training duration used.

As seen in Table 3, Model 1 demonstrated the best overall balance between validation performance and test generalization. The model achieved over 82% accuracy while maintaining strong F1-score across all seven vibe categories.

6.1 Training Results Across Experiments

Model	Accuracy	Precision	Recall	Weighted F1
Model 1	0.8267	0.8335	0.8267	0.8265
Model 2	0.7633	0.7598	0.7633	0.7591
Model 3	0.8233	0.8251	0.8233	0.8228

Table 3: Final test performance across DistilBERT experiments

Figure 1 shows the training and validation loss across epochs for the final selected DistilBERT model.

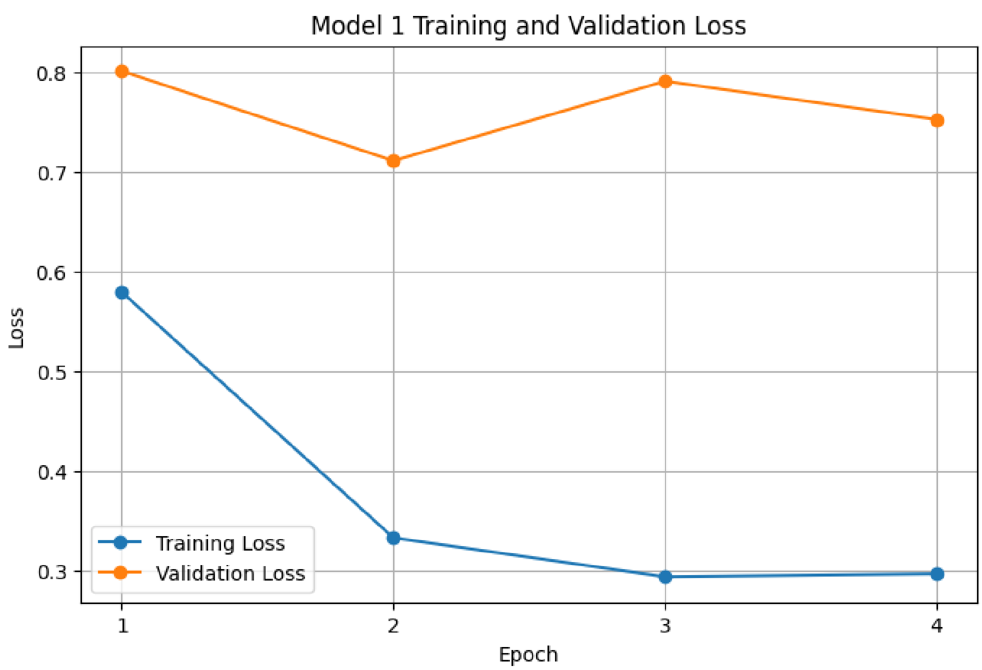

Figure 1: Training and validation loss curves for the final DistilBERT model

6.2 Final Model Comparison

The final DistilBERT model (Model 1) was also compared against a traditional NLP baseline using TF-IDF vectorization and Logistic Regression classification.

The larger 2,000-example dataset revealed a much clearer performance advantage for the transformer-based approach compared to the earlier 200-example experiments. DistilBERT significantly outperformed the TF-IDF baseline in both accuracy and weighted F1-score. This suggests that transformer models benefit substantially from larger datasets and are better able to learn contextual relationships between restaurant descriptions and atmosphere labels.

Model	Accuracy	Weighted F1
TF-IDF + Logistic Regression	0.70	0.70
DistilBERT (Model 1)	0.826	0.826

Table 4: Baseline vs. fine-tuned DistilBERT performance

6.3 Validation Performance During Training

Model 1 showed relatively stable learning behavior across all four training epochs. The model converged quickly within only a few epochs while maintaining relatively stable validation loss and strong overall classification performance.

Epoch	Train Loss	Val Loss	Accuracy	Precision	Recall	F1
1	0.5792	0.8005	0.7633	0.7872	0.7633	0.7656
2	0.3333	0.7107	0.8067	0.8167	0.8067	0.8079
3	0.2943	0.7903	0.8000	0.8015	0.8000	0.7997
4	0.2973	0.7521	0.8133	0.8170	0.8133	0.8141

Table 5: Model 1 validation performance during training

6.4 Example Predictions and Qualitative Analysis

The final DistilBERT model was generally successful at identifying contextual restaurant atmosphere or "vibe" patterns from Yelp reviews. Compared to the TF-IDF baseline, the transformer model handled semantically overlapping categories more effectively because it relied on broader contextual language understanding rather than simple keyword matching alone.

In particular, DistilBERT showed stronger performance when distinguishing between labels such as:

- *date_night* vs. *casual_hangout*
- *trendy* vs. *upscale*
- *study_spot* vs. *casual_hangout*

To better understand model behavior beyond aggregate metrics, several qualitative prediction examples were reviewed from the held-out test set. These examples include both successful classifications and semantically reasonable prediction errors.

Although some predictions did not exactly match the manually assigned ground-truth labels, many of the errors were still contextually reasonable. For example, a quiet coffee shop may reasonably function as both a *study_spot* and a *casual_hangout*, while stylish cocktail lounges may overlap naturally between *trendy* and *date_night* categories. These examples highlight one of the core challenges of restaurant vibe classification: atmosphere labels are often subjective, overlapping, and dependent on interpretation rather than strictly objective definitions.

6.5 Confusion Matrix

Figure 2 shows the confusion matrix for the final DistilBERT model evaluated on the held-out test set. The confusion matrix shows strong diagonal concentration across most vibe categories, indicating that the model learned meaningful separation between restaurant vibe labels.

Input Summary	True Label	Predicted Label
Quiet cafe with students studying and working on laptops	study_spot	casual_hangout
Stylish cocktail lounge with modern atmosphere and drinks	trendy	date_night
Upscale wine-focused restaurant with elegant dining	upscale	trendy
Family breakfast diner with large seating and relaxed atmosphere	family_friendly	family_friendly
Late-night bar with music and social atmosphere	late_night	late_night

Table 6: Example predictions from the final DistilBERT model

The strongest-performing classes include: *late_night*, *upscale*, *trendy*, and *family_friendly*. Other semantically similar labels contain some overlap such as *date_night* and *casual_hangout* or *study_spot* and *casual_hangout*. However, many of these errors were still semantically reasonable because restaurants often naturally fit multiple atmosphere categories simultaneously.

7 Error Analysis

Although the final DistilBERT model achieved strong overall performance, several classification errors still occurred between semantically overlapping restaurant vibe categories. Many of these mistakes were understandable because restaurant atmosphere labels are inherently subjective and often share similar contextual language.

For example, one of the most common sources of confusion involved the relationship between *trendy*, *date_night*, and *upscale* restaurants. Many reviews describing modern cocktail lounges, wine bars, or stylish restaurants contained language that could reasonably fit multiple labels simultaneously. Restaurants described as “modern,” “romantic,” or “great for drinks” could appear both *trendy* and *date_night* depending on interpretation. Because these labels naturally share contextual similarities, some predictions that were technically incorrect were still semantically reasonable. Another common overlap occurred between *study_spot* and *casual_hangout*. Coffee shops and cafes frequently serve multiple purposes and may function as both relaxed social spaces and quiet work environments. Reviews mentioning coffee, seating, and relaxed atmosphere sometimes lacked enough explicit information for completely unambiguous classification.

The confusion matrix also showed that some categories with broader definitions, such as *casual_hangout*, occasionally absorbed predictions from neighboring classes because the label itself overlaps naturally with many restaurant types. Overall, many model errors reflected the ambiguity of the labeling task itself rather than complete failures in language understanding. In several cases, the model predictions were still contextually defensible even when they did not exactly match the manually assigned ground-truth label. This highlights an important challenge in restaurant atmosphere classification: vibe categories are often subjective, overlapping, and dependent on personal interpretation rather than strictly objective definitions.

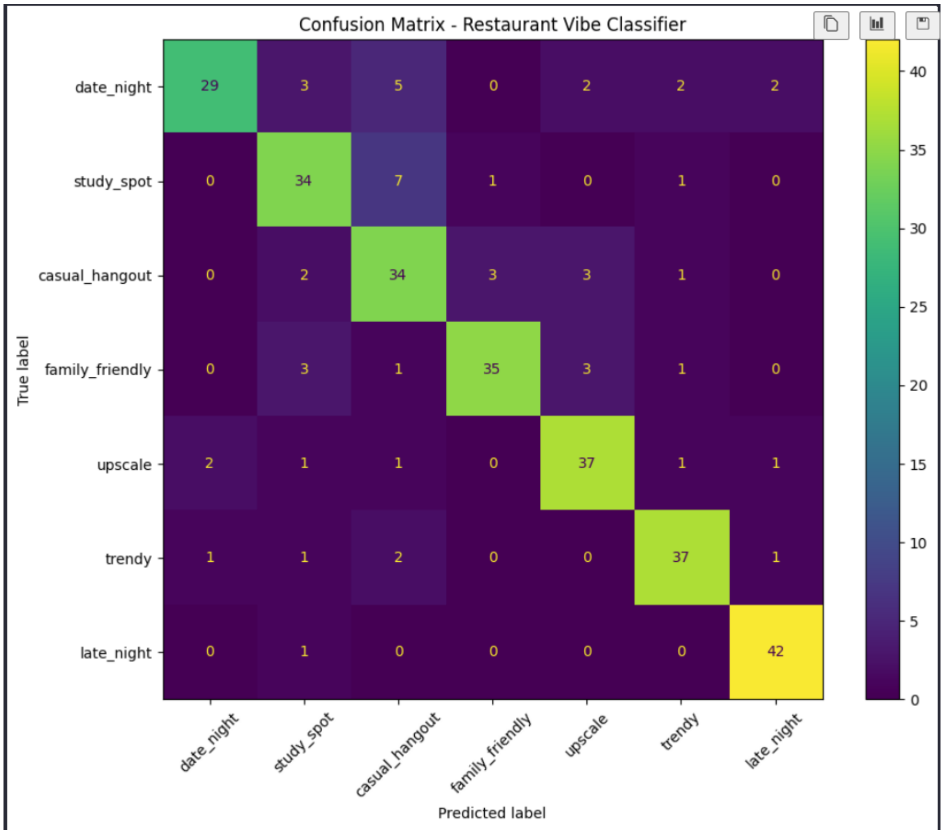

Figure 2: Confusion matrix for the final DistilBERT model

8 Ethical and Practical Considerations

One of the main challenges in this project is that restaurant "vibe" labels are inherently subjective. Different users may interpret the restaurant's atmosphere differently depending on personal preferences, dining expectations, or cultural context. For example, one user might consider a restaurant to be *trendy* while another user might view it as a nicer, more elegant, *upscale* location based on their personal backgrounds.

Because the dataset labels were initially manually labeled and then reviewed and supervised by one person, some level of labeling bias is unavoidable. Although efforts were made to remain impartial and maintain consistency and balanced in the labeling process, it is still reflective of some human interpretation. AI-assisted label recommendations were also used during dataset expansion, which may introduce additional bias or reinforce patterns already present in the data.

Another limitation involves geographic bias. The final dataset focused only on restaurants located in Santa Barbara, Goleta, and Carpinteria because of dataset availability constraints. Restaurant culture, dining trends, and review language may differ significantly across other cities or regions, limiting how well the model generalizes beyond the selected geographic area.

The data also inherits biases present within Yelp reviews themselves. Yelp users represent only a subset of restaurant customers, and review text may disproportionately reflect highly positive or highly negative dining experiences. Additionally, some restaurant types naturally receive more reviews and more descriptive language than others, which can influence model learning behavior.

Although the final 2,000-example dataset performed substantially better than the earlier 200-example experiments, the dataset size is still relatively small compared to large-scale NLP training datasets. Expanding the dataset further could improve generalization, reduce overlap confusion between vibe categories, and provide more robust evaluation across diverse restaurant types.

Despite these limitations, the project demonstrates that locally fine-tuned transformer models can still learn meaningful contextual atmosphere patterns from authentic restaurant review text while operating within realistic hardware and project constraints.

9 Conclusion

This project explored whether locally fine-tuned transformer models could classify restaurant atmosphere or "vibe" categories using Yelp review text and restaurant metadata. The primary goal was to build a domain-specific NLP system capable of identifying restaurant experiences such as *study_spot*, *date_night*, *trendy*, and *late_night* using natural language review data.

A custom preprocessing and labeling pipeline was developed using the Yelp Open Dataset after originally planned Orange County datasets proved inaccessible due to licensing costs. The final dataset consisted of approximately 2,000 balanced restaurant-review examples focused on the Santa Barbara region. Multiple DistilBERT modelts were trained, tested and compared locally against a traditional TF-IDF + Logistic Regression baseline.

The final DistilBERT model achieved approximately 83% accuracy and weighted F1-score, outperforming the traditional NLP baseline and demonstrating stronger contextual understanding of overlapping restaurant vibe categories. Confusion matrix analysis and qualitative examples showed that many remaining prediction errors were still semantically reasonable due to the inherently subjective nature of restaurant atmosphere classification. Overall, the project demonstrated that compact transformer models such as DistilBERT can perform effective domain-specific NLP classification tasks locally while remaining computationally feasible on consumer hardware.

Future improvements could include:

- expanding the labeled dataset further
- incorporating additional restaurant metadata
- experimenting with larger transformer architectures
- testing parameter-efficient fine-tuning methods such as LoRA or QLoRA
- building a full restaurant recommendation assistant on top of the classifier

A Appendix: Full Training Outputs

This appendix contains the full training outputs and evaluation metrics for all three DistilBERT fine-tuning experiments tested during the project.

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.579195	0.800463	0.763333	0.787189	0.763333	0.765551
2	0.333281	0.710718	0.806667	0.816737	0.806667	0.807914
3	0.294274	0.790323	0.800000	0.801515	0.800000	0.799696
4	0.297294	0.752081	0.813333	0.817016	0.813333	0.814145

Table 7: Model 1 validation performance during training

Training Loss	Validation Loss	Epoch	Accuracy	Precision	Recall	F1
0.297294	0.684436	4	0.826667	0.833478	0.826667	0.826456

Table 8: Final test performance for Model 1

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.703151	0.941842	0.666667	0.683242	0.666667	0.661912
2	0.463085	0.823341	0.723333	0.733014	0.723333	0.724832
3	0.398271	0.816910	0.756667	0.763174	0.756667	0.757411
4	0.362519	0.794781	0.770000	0.782044	0.770000	0.771454

Table 9: Model 2 validation performance during training

Training Loss	Validation Loss	Epoch	Accuracy	Precision	Recall	F1
0.362519	0.761202	4	0.763333	0.759801	0.763333	0.759082

Table 10: Final test performance for Model 2

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.648392	0.846117	0.736667	0.752028	0.736667	0.738996
2	0.390244	0.774241	0.786667	0.795410	0.786667	0.788210
3	0.327918	0.745102	0.816667	0.821315	0.816667	0.817602
4	0.301102	0.721993	0.843333	0.849271	0.843333	0.844292

Table 11: Model 3 validation performance during training

Training Loss	Validation Loss	Epoch	Accuracy	Precision	Recall	F1
0.301102	0.701488	4	0.823333	0.825140	0.823333	0.822761

Table 12: Final test performance for Model 3